

Growth Opportunities Connected With The Growing SEMICON ECOSYSTEM IN INDIA

India must concentrate on electronics products and semiconductor components, as progress in one area reinforces the other. The country can unlock immense economic opportunities and enhance its global standing by aiming to capture a significant market share in these areas



DR SATYA GUPTA,
President, VLSI
Society of India,
Co-Chair, FICCI
Digital Economy
Committee

A country's identity is determined by the products that the country creates, and technical progress is intrinsically linked to the innovative products that the country brings to the market. When we think of such a country, one name that often comes to mind is Japan. Products like Sony, Toyota, and Toshiba are synonymous with Japan's identity. The types of products a country creates fundamentally define the destiny of a country and its place on the world map.

In the modern era, electronics and semiconductors have emerged as the lifeline for everyone. We use hundreds of semiconductor chips daily, driving almost every electronics product. From the moment we wake up to the car we drive, the computer we use, the mobile devices that keep us connected, movies and music we hear, electronics and semiconductor devices that drive them play an omnipresent role. These technologies are not limited to personal use - they permeate every sector, including education, mobility, healthcare, infrastructure, defence, space, etc. Electronics

and semiconductors have become the foundation of our modern world and economies.

Transformative impact of semiconductor on everything

Semiconductors have enabled most of the contemporary age technologies like 5G, IoT, AI, and others. AI has experienced unprecedented popularity and success in recent years and is considered the most critical technology of human evolution. We find AI applications in everything, from assisting us in crafting the perfect resume to solving complex problems in science & engineering. Generative AI today is transforming every industry. Although foundational AI algorithms such as 'backpropagation' and neural networks have existed for many years, they have not taken off because of limited computing and storage capacity. Simulating a simple neural network with just 500 nodes required a tiny supercomputer. However, the landscape has transformed dramatically over the years. The exponential growth in